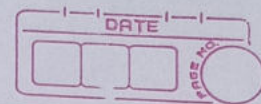


VECTOR DIFFERENTIATION



$$1) \frac{d}{dt} A \cdot B = A \frac{dB}{dt} + B \cdot \frac{dA}{dt}$$

$$2) \frac{d}{dt} (A \times B) = A \times \frac{dB}{dt} + \frac{dA}{dt} \times B$$

$$Q. A = 5t^2 \hat{i} + t \hat{j} - t^3 \hat{k}$$

$$B = \sin t \hat{i} - \cos t \hat{j}$$

$$\Rightarrow \frac{d}{dt} A \cdot B \Rightarrow A \frac{dB}{dt} + B \cdot \frac{dA}{dt}$$

$$= (5t^2 \hat{i} + t \hat{j} - t^3 \hat{k}) \cdot (\cos t \hat{i} + \sin t \hat{j}) + (\sin t \hat{i} - \cos t \hat{j}) \cdot (10t \hat{i} + \hat{j} - 3t^2 \hat{k})$$

$$= 5t^2 \cos t + t \sin t + 10t \sin t - \cos t$$

$$= 5t^2 \cos t - \cos t + 11t \sin t \quad \text{Ans.}$$

$$\Rightarrow \frac{d}{dt} A \times B \Rightarrow A \times \frac{dB}{dt} + \frac{dA}{dt} \times B \Rightarrow A \times (\cos t \hat{i} + \sin t \hat{j}) + (10t \hat{i} + \hat{j} - 3t^2 \hat{k}) \times B$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5t^2 & t & -t^3 \\ \cos t & \sin t & 0 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 10t & 1 & -3t^2 \\ \sin t & -\cos t & 0 \end{vmatrix}$$

$$\Rightarrow \hat{i} [0 + t^3 \sin t] - \hat{j} [0 + t^3 \cos t] + \hat{k} [5t^2 \sin t - t \cos t]$$

$$+ \hat{i} [0 - 3t^2 \cos t] - \hat{j} [0 + 3t^2 \sin t] + \hat{k} [-10t \cos t - \sin t]$$

$$\Rightarrow \hat{i} [t^3 \sin t - 3t^2 \cos t] - \hat{j} [t^3 \cos t + 3t^2 \sin t] + \hat{k} [5t^2 \sin t - t \cos t - 10t \cos t - \sin t]$$

Ans

* gradient of f (grad f or ∇f)
 The vector function ∇f is define as the gradient of scalar point function f and it written as grad f
 thus

$$\text{grad } f = \nabla f = i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z}$$

$$\text{normal vector} = \text{grad } f = \nabla f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) f$$

Q $f = xy^3z^2 - 4$

$$\Rightarrow \text{normal vector} = \text{grad } f = \nabla f = i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z}$$

$$\nabla f = i(y^3z^2) + j(xz^2 \times 3y^2) + k(xy^3 \times 2z)$$

$$\nabla f = y^3z^2 \hat{i} + 3xy^2z^2 \hat{j} + 2xy^3z \hat{k} \quad \text{Ans}$$

$$\Rightarrow \text{unit normal vector} = \frac{\text{grad } f}{|\text{grad } f|}$$

at point $(-1, -1, 2)$

$$= \frac{y^3z^2 \hat{i} + 3xy^2z^2 \hat{j} + 2xy^3z \hat{k}}{\sqrt{(y^3z^2)^2 + (3xy^2z^2)^2 + (2xy^3z)^2}}$$

$$= \frac{-4 \hat{i} - 12 \hat{j} + 4 \hat{k}}{\sqrt{(4)^2 + (12)^2 + (16)^2}}$$

$$= \frac{4(-\hat{i} - 3\hat{j} + \hat{k})}{4(\sqrt{1+9+1})}$$

$$= \boxed{\frac{-\hat{i} - 3\hat{j} + \hat{k}}{\sqrt{11}}} \quad \text{Ans.}$$

* Directional Derivative = $(\text{grad } f)_p \frac{\vec{N}}{|\vec{N}|}$

$\vec{N}$ = direction of vector

ex- $f(x, y, z) = xy^2 + yz^3$

at point $(2, -1, 1)$

direction of vector $\vec{N} = \hat{i} + 2\hat{j} + 2\hat{k}$

$$\begin{aligned} \Rightarrow (\text{grad } f)_p &= \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} \\ &= \hat{i}(y^2) + \hat{j}(2xy + z^3) + \hat{k}(3z^2 y) \\ &\quad \hat{i}(-1)^2 + \hat{j}(2(2)(-1) + 1) + \hat{k}(3(1)^2(-1)) \\ &= \hat{i}(1) + \hat{j}(-4 + 1) + \hat{k}(-3) \\ &= \hat{i} - 3\hat{j} - 3\hat{k} \text{ Ans.} \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Directional Derivative} &= (\text{grad } f)_p \frac{\vec{N}}{|\vec{N}|} \\ &= \frac{(\hat{i} - 3\hat{j} - 3\hat{k}) \cdot (\hat{i} + 2\hat{j} + 2\hat{k})}{\sqrt{1+4+4}} \\ &= \frac{1 - 6 - 6}{3} \\ &= \frac{-11}{3} \text{ Ans.} \end{aligned}$$

* Divergence = The divergence of continuously differential vector point function F , it denoted by $[\text{div } F = \nabla \cdot F]$
it is define by equation

$$\nabla \cdot F = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot F$$

$$F = f\hat{i} + \phi\hat{j} + \psi\hat{k}$$

$$\nabla \cdot F = \frac{\partial f}{\partial x} + \frac{\partial \phi}{\partial y} + \frac{\partial \psi}{\partial z}$$

Q. $u = x^3 + y^3 + z^3 - 3xyz$

$$\Rightarrow F = \text{grad} f = \nabla f = \hat{i} \frac{\partial u}{\partial x} + \hat{j} \frac{\partial u}{\partial y} + \hat{k} \frac{\partial u}{\partial z}$$

$$= \hat{i}(3x^2 - 3yz) + \hat{j}(3y^2 - 3xz) + \hat{k}(3z^2 - 3xy)$$

$$\Rightarrow \text{div} F = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot F$$

$$= 3 \frac{\partial}{\partial x} x^2 + 3 \frac{\partial}{\partial y} y^2 + 3 \frac{\partial}{\partial z} z^2$$

$$= 6x + 6y + 6z \quad \text{Ans.}$$

* $\text{curl} F = \nabla \times F = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times F$